

RealView® Microcontroller Development Tools

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Director of MCU Tools



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About This Presentation

- Introduction and Overview
 - Keil™ – An ARM Company
 - Positioning of RealView Microcontroller Development Kit
- RealView Microcontroller Development Kit
 - µVision® Integrated Development Environment
 - RealView compilation tools
 - Hardware components
 - ULINK™-2 USB to JTAG adapter
 - Evaluation boards
- RealView Real-Time Library
- Business Model



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Keil – An ARM Company

- Leading supplier of MCU development tools
 - ANSI C/C++ compilers, debuggers
 - Device simulators
 - Real-time kernels, real-time libraries
- Extensive Device Database®
 - Including >1,100 8/16/32-bit MCUs
- Excellent, established support
 - Web support portal
 - User group structure
 - Global distribution network
- Huge installed base
 - 100K+ users worldwide



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Introduction

RealView Microcontroller Development Kit



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SoC and MCU Require Different Tools

- | RealView Development Suite | RealView Microcontroller Development Kit |
|---|---|
| <ul style="list-style-type: none"> Market <ul style="list-style-type: none"> Creators of SoC/ASIC/FPGA devices Complex embedded application developers Care about <ul style="list-style-type: none"> ARM processor IP centric support Multiple ARM processor+DSP architecture support Flexible system support for custom SoC SoC bring-up Advanced debug and analysis features Platform OS and RTOS support Early IP adopter support Interface into RealView CREATE (ESL) world Do not care about <ul style="list-style-type: none"> Very little—want it all | <ul style="list-style-type: none"> Market <ul style="list-style-type: none"> Users of MCU devices Typical applications are straightforward, focused on control or sampling data Care about <ul style="list-style-type: none"> MCU-specific debug and simulation Ease of use, no redundant features Suitable for generalist engineers Deeply embedded systems only Low to mid price point, free evaluation Off-the-shelf firmware and kernels Do not care about <ul style="list-style-type: none"> Multi-core debug Platform Operating Systems Interface to RealView CREATE (ESL) world |

One size cannot fit all

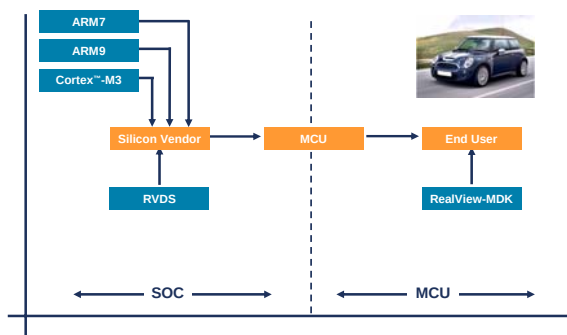


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MCU Development Process



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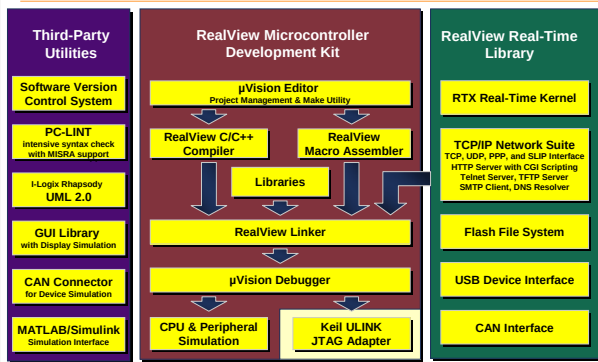


RealView Microcontroller Development Kit

- Industry-leading technology
 - RealView compilation tools
 - Keil μ Vision Integrated Development Environment
- Supports devices
 - ARM7[™], ARM9[™] and Cortex-M3 processor-based MCUs
 - Start-up code, Flash algorithm etc.
 - Complete device simulation
- Optional hardware products
 - ULINK USB to JTAG adapter
 - Evaluation boards
- RTX real-time kernel
 - Efficient RTOS kernel for small systems



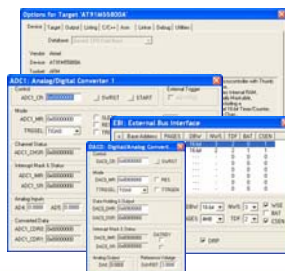
RealView Microcontroller Development Kit



Microcontroller Support

Instantly begin developing software, without the need for hardware or in-depth knowledge of target MCU

- Out-of-the-box support
 - >150 ARM MCUs
 - See www.keil.com/dd
- Complete simulation of target
 - Simulates entire target hardware
 - Including external signals and I/O
 - Fast Instruction Set Simulation
 - Interrupt simulation
 - On-chip peripherals
 - ADC, DAC, EBI, Timers
 - UART, CAN, I2C...



Benefits

- Benefits for end users – lead to faster MCU adoption
- End users
 - Ease of use
 - Features that accelerate MCU-based development
 - Quick target device configuration
 - Pay for what they need – no redundant features
- Silicon Partners
 - End users comfortable with development environment
 - Features of RealView MDK move design to production faster
 - Comfortable to move to new MCUs supported in RealView MDK
 - Volume silicon ships sooner



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Ease of Use

RealView Microcontroller Development Kit



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4 Steps

RealView Microcontroller Development Kit



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Create Program

Step 1: Select Device and Specify Target Hardware



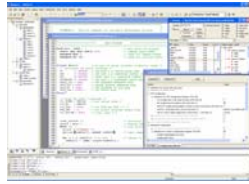
Device Database on Web and μ Vision simplifies chip selection and setup



Step 2: Configure Device and Create Application Code



μ Vision includes project manager and editor and debugger



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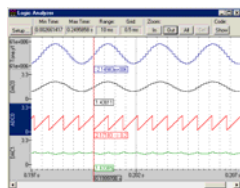


Program Testing

Step 3: Analyze Program with μ Vision Device Simulation



Debugging via μ Vision debugger and device simulator



Step 4: Flash Download and Final Testing in Target Hardware



Flash download and target debugging via ULINK



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μ Vision Integrated Development Environment RealView Microcontroller Development Kit



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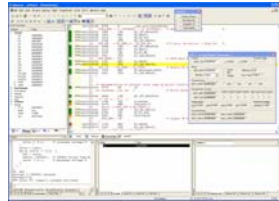


µVision IDE



Leading microcontroller Integrated Development Environment

- Common to ARM, C166 and 8051 platforms
- Intuitive project management
 - Including target device configuration and Device Database
- High-speed simulation
 - Instruction set
 - On-chip peripherals
- Source code editor
- Project debugging
- Flash programming
- Intelligent make functions



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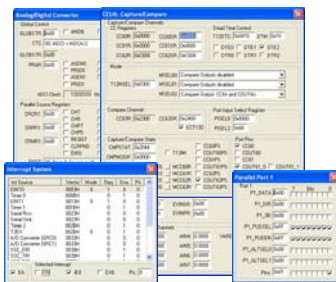


µVision Device Simulation



Simulation allows software testing with no hardware, providing quick test cycles and improving product quality

- Complete simulation
 - Many standard devices
 - Covers complex peripherals: ADC, DAC, EBI, Timers...
 - for >150 ARM processor-based MCUs
- Peripheral dialogs
 - Simulation
 - JTAG debugging
- Expandable
 - User defined DLLs



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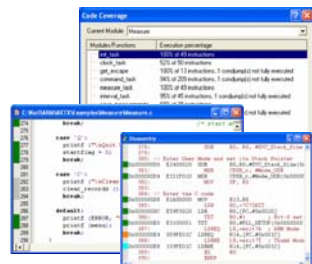
µVision Code Coverage



Simulation gives features not available with JTAG debuggers

Example: Code coverage at source and assembly level

- Execution statistics
 - Always active for complete project
 - Instruction status:
 - Not executed (grey)
 - Fully executed (green)
 - Skipped (orange)
 - Always taken (cyan)
 - Multi-session coverage with save / restore



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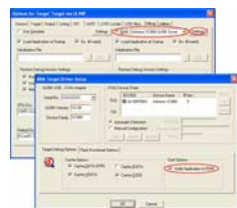
ARM

-
- | Channel | Waveform | Cursor | Data |
|---------|----------|------------|------------|
| 0 | 00000000 | 0.000000ms | 0.000000ms |
| 1 | 00000000 | 0.000000ms | 0.000000ms |
| 2 | 00000000 | 0.000000ms | 0.000000ms |



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RealView Compilation Tools

RealView Microcontroller Development Kit



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RealView Compilation Tools

Best-in-class tools, making code smaller and more efficient
Enabling applications to run faster and reduce system cost

- RealView MDK uses compilation tools from RealView Development Suite 3.0
 - Best-in-class compiler, linker, assembler
- The RealView compilation tools contain:
 - Highly-optimizing ISO C/C++ compiler
 - Supports ARM, Thumb® and Thumb-2 instruction sets
 - Full C and C++ run-time library support
 - RogueWave C++ standard template libraries
- The RealView advantages:
 - Smaller, faster code



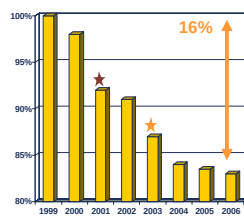
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RealView Compilation Improvement

Code Size Reduction

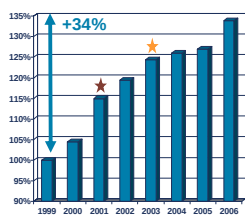


SDT 2.5 → RVCT 3.0

9MB ROM Size

■ 46 Benchmarks, 48 Applications

Performance Increase



SDT 2.5 → RVCT 3.0

★ ADS v1.2

★ RealView 2.0



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RealView Real-Time Library

RealView Microcontroller Development Kit



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RealView Real-Time Library

Extensive library of common ready-to-use components speeds software development

RealView Real-Time Library

RTX Real-Time Kernel

TCP/IP Network Suite

TCP, UDP, PPP, and SLIP Interface
HTTP Server with CGI Scripting
Telnet Server, TFTP Server
SMTP Client, DNS Resolver

Flash File System

USB Device Interface

CAN Interface

- Meets embedded developers' needs
 - Solves common embedded challenges
 - Real-time systems
 - Embedded communication
 - Designed for use with MCU devices
 - High level of re-use
- Extensive range of application examples
 - Embedded control, networking, HTTP...
- Royalty free
 - RTX (object) included in RealView MDK



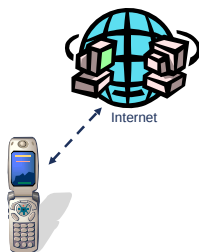
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RTX Real-Time Kernel

- Multitasking
 - Manage several tasks on single CPU
- Real-time control
 - Execution of tasks within defined time period
- Inter-task communication
 - Effective communication within system
- Internet connectivity
 - Via Ethernet, modem, GSM...
- Embedded Web server
 - Easy hosting and interface of HTML information
- E-mail notification
 - Via SMTP – ideal for remote applications



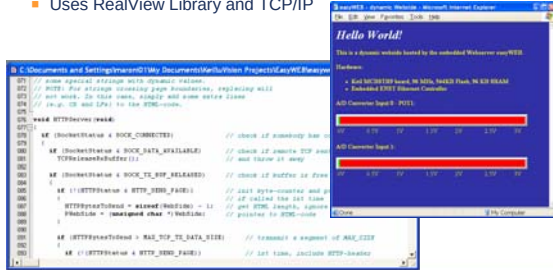
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RealView Real-Time Library Examples

- easyWeb
 - Easy web interface to examine two I/O levels
 - Ideal for remote data monitoring applications
 - Uses RealView Library and TCP/IP



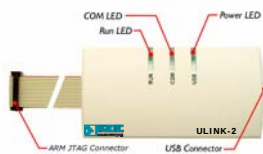
Hardware

RealView Microcontroller Development Kit

ULINK-2 USB to JTAG Adapter

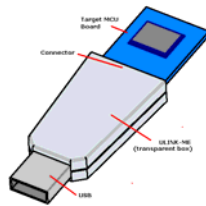
Easy-to-use USB run control adapter, supporting JTAG, serial wire debug and Flash programming

- New features in ULINK-2
 - Cortex-M3 serial wire debug and trace
 - Higher speed – 100KB/Sec
- Program debugging
 - Flash and software breakpoints
 - Memory access breakpoints
 - Real-time execution
- Flash programming
 - Download, verify, and execute
 - Customized programming algorithms



ULINK-Me Concept

- Extremely low cost
 - <\$70 end customer price
- Development kit on a stick
 - USB debug adapter
 - JTAG and SWD/SVV
 - Interchangeable target board
 - Easy evaluation of product families
- Small size
 - Flash memory stick size
 - 77 x 24mm total



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ARM

Wide Range of Evaluation Boards

Proven hardware for quick project development and debug

- Designed for easy set-up
 - Extensive program examples
 - Wide range of peripherals
 - Popular ARM processor-based MCUs

STR91x



LPC2148



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ARM

RealView Microcontroller Development Tools



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ARM

Get More Information

Customers use www.keil.com on a daily basis to obtain program examples, latest technical information, and support

- Application notes
- Program examples
- Device database
- Support knowledge
- Discussion forum



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